



ICRAF Spatial Data Science and Applied Learning Lab (SPACIAL)

AARAN, Somalia

Land Restoration Prioritization Report

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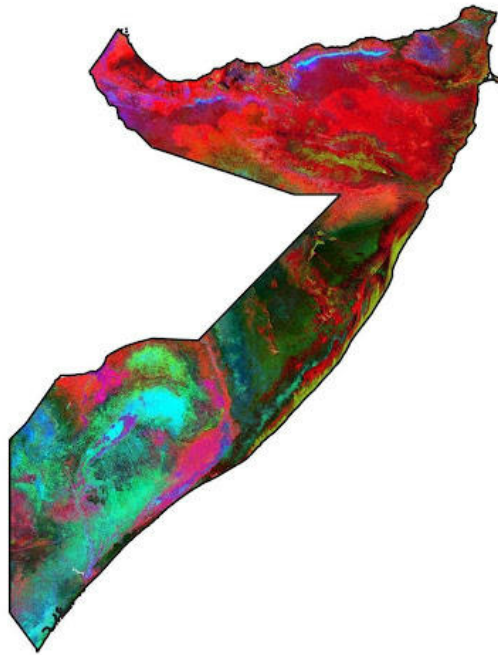


Table of contents

1 Introduction	3
2 Methods	3
2.1 Study area and sampling design	3
2.2 Biophysical indicators	3
2.3 Model-based clustering	4
2.4 Interpreting clusters as restoration effort classes	4
2.5 Probability surfaces and variable importance	4
3 Results	5
4 Random Forest Results	6
4.1 Variable importance	7
5 Country-Level Overview	7
6 Watershed and Cluster-Level Analysis	10
6.1 Watershed 1 and KAALO_6	11
6.2 Watershed 2 and KAALO_1	13
6.3 Watershed 3 and CWW_4	16
6.4 Watersheds 5 and 6 with GREDO_6 and GREDO_7	19
6.5 Watershed 7 and CWW_8	23
6.6 Watershed 8 and KAALO_7	26
7 Restoration Recommendations	29
8 Conclusion	30
9 Bibliography	31

1 Introduction

This report presents the results of land health analysis and assessment of restoration effort to support the prioritisation of land restoration interventions in Somalia for the AARAN project. Restoration effort is assessed using soil and land health indicators, which are clustered into low, medium, and high restoration effort, respectively. The classes are derived from a model-based clustering workflow using five biophysical indicators: tree cover, erosion, root depth restriction at 50 cm, soil pH, and soil organic carbon (SOC).

Throughout the report, restoration effort classes are also interpreted in terms of level of concern for land degradation. Areas classified as high restoration effort correspond to high concern, as they combine indicators associated with active or severe degradation, such as high erosion, low SOC, and unfavourable soil conditions. Conversely, areas of low restoration effort reflect comparatively healthy land and are of low concern. Medium effort areas fall between these extremes. Framing the classes in this way makes clear that the restoration effort gradient is simultaneously a gradient of degradation concern: where intervention needs (effort) are greatest, the underlying land health condition gives greatest cause for concern.

The purpose of the report is to give an overview at country, watershed and cluster level, and to provide a summary of the indicator values associated with each restoration effort class. The report is intended for use by AARAN and other stakeholders to inform restoration planning and prioritization.

i Key Findings

- **National Effort Distribution:** Approximately **37%** Medium restoration effort, **34%** High restoration effort, and **29%** Low restoration effort.
- **Biophysical Clustering:** Five indicators (tree cover, erosion, RDR50, soil pH, and SOC) grouped into 3 distinct operational management classes via Gaussian mixture modeling (`mclust`).
- **Primary Predictive Drivers:** Tree cover and soil pH consistently show the highest predictive influence across class-specific Random Forest probability models.

2 Methods

2.1 Study area and sampling design

The assessment covers the whole of Somalia. To obtain a representative sample of the biophysical indicator values across the country, points were generated at random locations and the values of five indicators were extracted for each point from the raster layers described below. This sampling approach ensures that the full range of environmental conditions present in Somalia is captured, including areas that are remote or otherwise underrepresented in field campaigns.

2.2 Biophysical indicators

The indicators used in this assessment are drawn from the Land Degradation Surveillance Framework (LDSF), a standardized field method developed by ICRAF (CIFOR-ICRAF) for assessing soil and land health (Vågen & Winowiecki, 2013; Winowiecki et al., 2021). The LDSF has been applied at over 55,000 sampling locations in the tropics over the past two decades, producing one of the largest consistent databases of ground observations on land health. Because the indicators are measured in the same way everywhere, they can be compared across sites and over time, a bit like a standardized health check-up, but for land rather than for people.

A total of about 1,100 sampling locations from 7 LDSF sites in Somalia were also part of the underlying data used to assess soil and land health in AARAN, and to conduct the analysis of restoration effort for the different clusters of the project. These data were collected as part of LDSF surveys in 2024, 2025, and 2026. Soil data was collected from one of these sites (Guriel) and analysed in Nairobi, Kenya.

Five LDSF-based indicators were used, each representing a key aspect of soil and land health:

1. **Tree cover:** the proportion of area covered by woody vegetation, an indicator of vegetation density and habitat provision.
2. **Erosion:** the prevalence or severity of soil erosion, a direct measure of land degradation.
3. **Root depth restriction at 50 cm (RDR50):** a measure of physical limitations to root development, reflecting soil depth and compaction constraints.
4. **Soil pH:** an indicator of soil chemical conditions, with values far from neutral generally limiting nutrient availability and plant productivity.
5. **Soil organic carbon (SOC):** a core indicator of soil fertility, water-holding capacity, and carbon storage.

2.3 Model-based clustering

Sampled indicator values were standardized prior to analysis and clustered into three groups using model-based clustering implemented in the `mclust` library in R Statistics (Scrucca et al., 2023). Unlike distance-based methods such as k-means, Gaussian mixture models fit a probabilistic model to the data and assign observations to clusters based on the posterior probability of membership. The number of clusters was set to three to match the operational need for low, medium, and high restoration effort classes.

2.4 Interpreting clusters as restoration effort classes

The three resulting clusters were manually interpreted as restoration effort classes based on the distribution of indicator values in each cluster. The cluster with high erosion, low SOC, and acidic or alkaline pH was classified as high restoration effort (high concern), while the cluster with low erosion, higher SOC, and near-neutral pH was classified as low restoration effort (low concern). The cluster with intermediate indicator values was classified as medium restoration effort (medium concern). Interpretation was supported by the indicator distributions shown in the Results section.

The following considerations were taken into account when interpreting the clusters:

- Areas with low SOC and high erosion are considered to be highly degraded and may require considerable resources and efforts to restore. Hence, active restoration will be required such as the establishment of soil and water conservation measures (e.g. halfmoons or terracing) combined with tree planting and/or grass re-seeding.
- Alkaline soils (pH > 7.5) can be high in SOC due to high clay content, but may also be challenging to restore when degraded.
- High tree cover in combination with low SOC and high erosion, can indicate invasive species which may require removal.
- High historic SOC concentrations indicate a high potential for carbon storage. Such areas generally also have a higher level of inherent resilience and may require less effort to restore.
- RDR50 maps where roots are physically restricted within the top 0.5 m of soil, indicating reduced access to water and nutrients and therefore lower potential vegetation growth. It helps decide whether to favor shallow-rooted species, invest in breaking hardpans, or limit expectations for deep-rooted vegetation in a given area.

2.5 Probability surfaces and variable importance

In order to scale the analysis to the country level, indicator maps were produced by using predictive models that learn the relationship between the LDSF ground observations (stored in a global database of field data from thousands of sites worldwide) and freely available satellite and environmental data, such as climate, terrain, and spectral information (Sanchez et al., 2009; Vågen et al., 2016). Once a model has “learned” these relationships, it can estimate the indicator values for every location in the country. We produced these maps at a spatial resolution of 250 m for the purpose of this exercise, but more detailed maps can also be generated using this approach. The reliability of this approach rests on the size and quality of the underlying ground truth database, which for the LDSF is the global tropics.

For each restoration effort class, a random forest classification model was developed using the sampled indicator values as predictors of class membership. The models were applied to the full indicator raster stack to produce spatially explicit probability surfaces, expressing for each pixel the probability of belonging to the low, medium, or high restoration effort class. Variable importance measures from the random forest models were used to identify which indicators most strongly distinguish each class, providing insight into the drivers of the observed patterns. Together, the class maps and probability surfaces support spatially explicit prioritization of restoration interventions across Somalia.

3 Results

Our analysis shows that SOC is low overall (median SOC below 10 g/kg), but with some sampled locations with SOC higher than 20 g/kg (Figure 1). Erosion is high overall (median > 60%), tree cover is highly variable, while pH ranges mostly from neutral to alkaline, many parts of the country having pH > 7.5. The Mclust model produced three restoration-effort groups, as described in Section 2.

1. *Low concern*: This class is characterized by moderate tree cover (mean 46%, median 47%), comparatively low erosion prevalence (mean 51%, median 49%), the highest SOC content of the three classes (mean 8.7, median 7.8 g/kg), near-neutral soil conditions (mean pH 7.1) and moderate RDR50 values (~47%). Overall, the relatively high vegetation cover, low erosion, and high soil organic carbon indicate comparatively favourable ecosystem conditions.
2. *Medium concern*: This class is characterized by higher tree cover (mean 59%, median 58%), moderate erosion prevalence (mean 61%, median 61%), moderate SOC content (mean 6.8, median 6.7 g/kg), near-neutral to slightly alkaline pH (mean 6.8) and intermediate RDR50 values (mean 56%). The class represents mixed ecosystem conditions, with high vegetation cover and near-neutral soils, but lower SOC than the low-concern class.
3. *High concern*: This class is characterized by very low tree cover (mean 24%, median 21%), high erosion prevalence (mean 70%, median 70%), very low SOC content (mean 3.7, median 3.6 g/kg), alkaline soil conditions (mean pH 7.5) and relatively low RDR50 values (~49%). The combination of sparse vegetation, high erosion, low soil organic carbon and alkaline soils indicates the strongest signs of ecosystem degradation among the three classes and therefore the greatest restoration concern. The very low SOC values mean that productivity is low and ecosystem services are impaired.

Throughout the report, the restoration effort classes are interpreted in terms of level of concern for land degradation. Areas classified as high restoration effort correspond to high concern, as they combine indicators associated with active or severe degradation, such as high erosion, low SOC, and unfavourable soil conditions (e.g. high pH). Conversely, areas of low restoration effort reflect comparatively healthy land and are of low concern, while medium effort areas fall between these extremes. The restoration effort gradient is therefore simultaneously a gradient of degradation concern: where intervention needs (effort) are greatest, the underlying land health condition gives the greatest cause for concern.

Areas of high concern have high levels of land degradation (erosion) and low SOC, often in combination with sparse tree cover and alkaline soils. These are the locations where restoration interventions will require the greatest investment, but also where the expected gains in land health and productivity are likely to be largest. Areas of low concern largely coincide with regions of higher tree cover and SOC, and can serve as reference or benchmark areas for restoration elsewhere in the country. Medium concern areas are transitional: they are generally still productive but show early warning signs of degradation, and are where relatively low-cost interventions may prevent further decline.

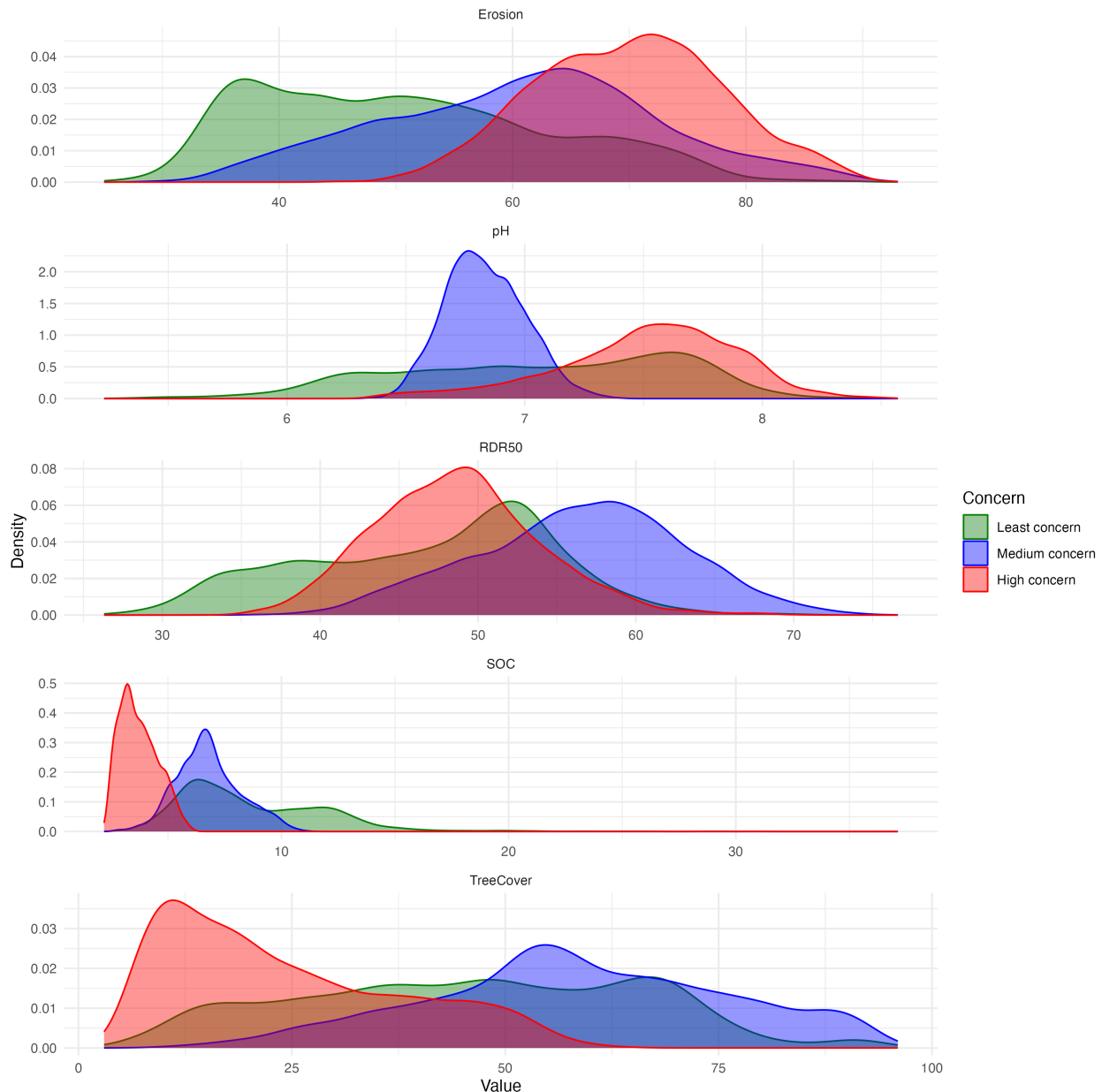


Figure 1: Graph showing the distribution of indicator values for each cluster across the randomly sampled points used to generate land restoration effort clusters

As we see in Figure 1, areas with high concern in terms of land degradation have high erosion (the red density graph is skewed to the right) while areas with low concern have low erosion (the blue density graph is skewed to the left). If we look at pH, we see that areas with high concern have high pH, while these areas have very low SOC (the blue density graph is skewed to the left in the SOC panel in Figure 1). For tree cover (bottom panel), the picture is a bit more mixed, but areas with high concern generally have very low tree cover. These groups provide the main classification used in the report, which we use to classify restoration effort within and across AARAN clusters.

4 Random Forest Results

A random forest model was used to predict low, medium, and high restoration effort (Breiman, 2001). It produces a class-specific probability surface for each class, and variable importance is assessed with class-specific (one-vs-rest) models to show which indicators contribute most to each class prediction.

4.1 Variable importance

Variable importance measures show how useful each indicator is for predicting a given restoration effort class. Random forests work by repeatedly splitting the data on the indicator values; at each split, an indicator that cleanly separates the classes reduces the uncertainty (“impurity”) of the classification. Importance is calculated as the total reduction in impurity achieved by each indicator across all splits in the model: the larger the reduction, the more the model relies on that indicator. The graphs in Figure 2 show this importance for each class, with longer bars indicating indicators that contribute more to the class prediction.

The results indicate that SOC is a strong predictor of high restoration effort, reflecting the distribution of SOC values in Figure 1 where we see a strong separation of SOC values between high and medium/low effort areas. Erosion is high overall with overlapping distributions in Figure 1.

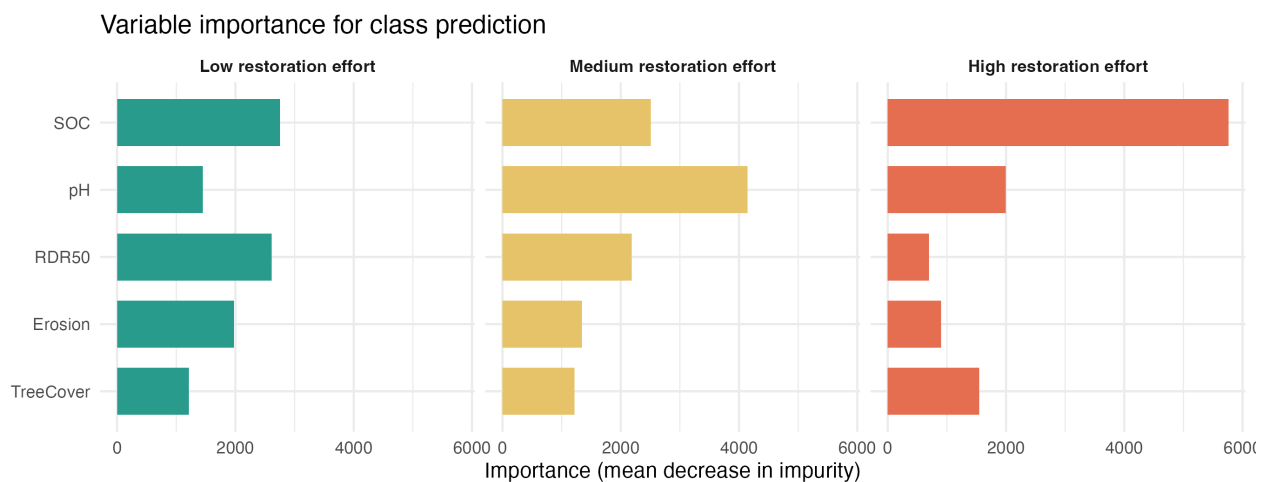


Figure 2: Variable importance for each restoration effort class, based on class-specific (one-vs-rest) random forest models

5 Country-Level Overview

At country level, the restoration-effort map shows the spatial distribution of low, medium, and high effort areas. The map is derived from the random forest probability surfaces generated at 250 m resolution for the whole country by assigning each pixel to the class with the highest predicted probability. The area graph summarizes how many square kilometers fall within each effort class, excluding hyperarid parts of the country.

The map reveals clear spatial patterns in restoration effort across Somalia. High effort areas, representing the greatest concern, are concentrated in the central and northern pastoral rangelands, where low SOC, high erosion prevalence, and alkaline soils coincide with sparse tree cover. Medium effort areas dominate the southern regions and transitional zones between the agro-pastoral belt and the rangelands, reflecting moderate land health conditions but with signs of degradation. Low effort areas are mainly found in the northwestern highlands and along the southern riverine and agroecological zones, where comparatively higher tree cover and SOC indicate better land health. These patterns provide an entry point for prioritizing restoration interventions.

⚠ About 71% of the mapped area (408,155 km²)

is classified as medium to high land degradation concern and will therefore require significant investment in restoration efforts.

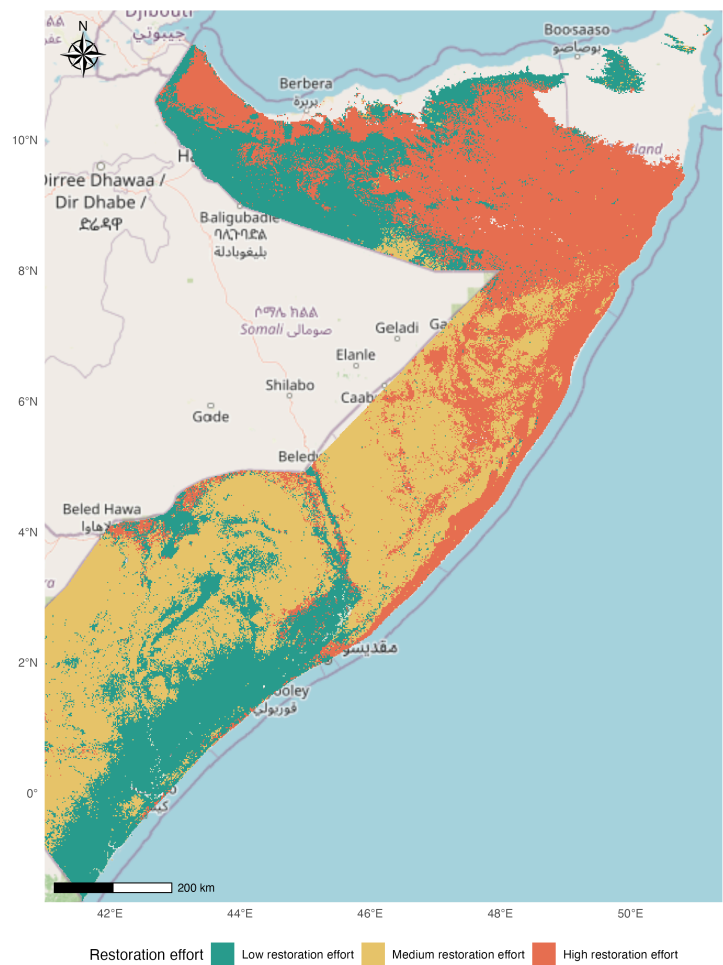


Figure 3: Map showing the spatial distribution of land degradation concern (restoration effort) classes for Somalia, derived from the highest-probability class in the random forest probability stack.

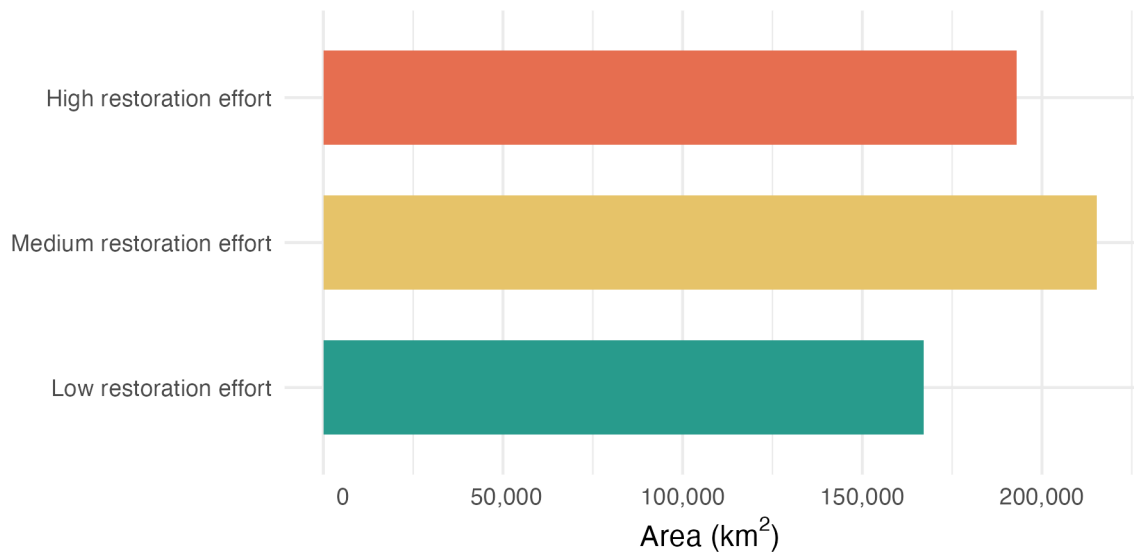


Figure 4: Total area per restoration effort class.

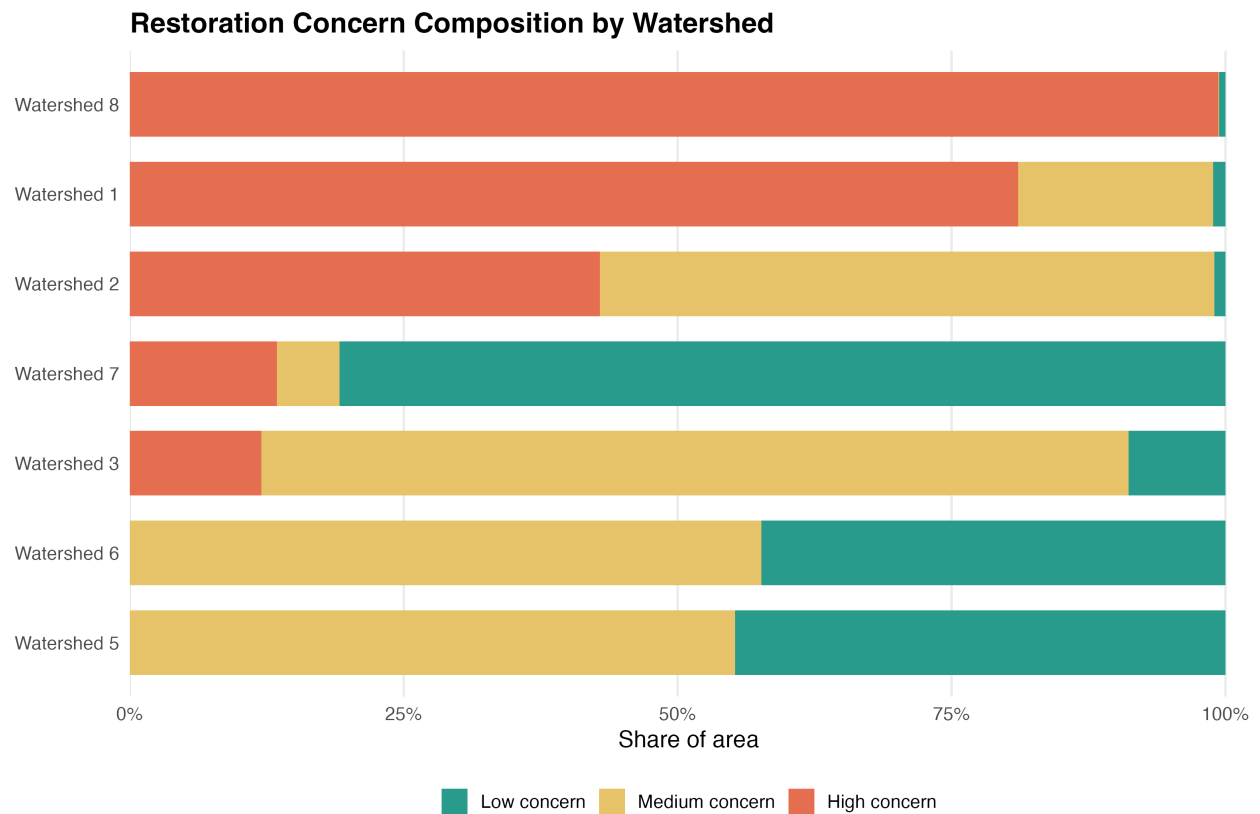


Figure 5: Share of restoration concern classes by watershed.

The figure above shows the distribution of the three concern classes across the watersheds. The results show that Watershed 8 in the Nugaal region has the highest percentage of area under high concern, while Watersheds 6 and 5 in Bay region barely have any area under high concern.

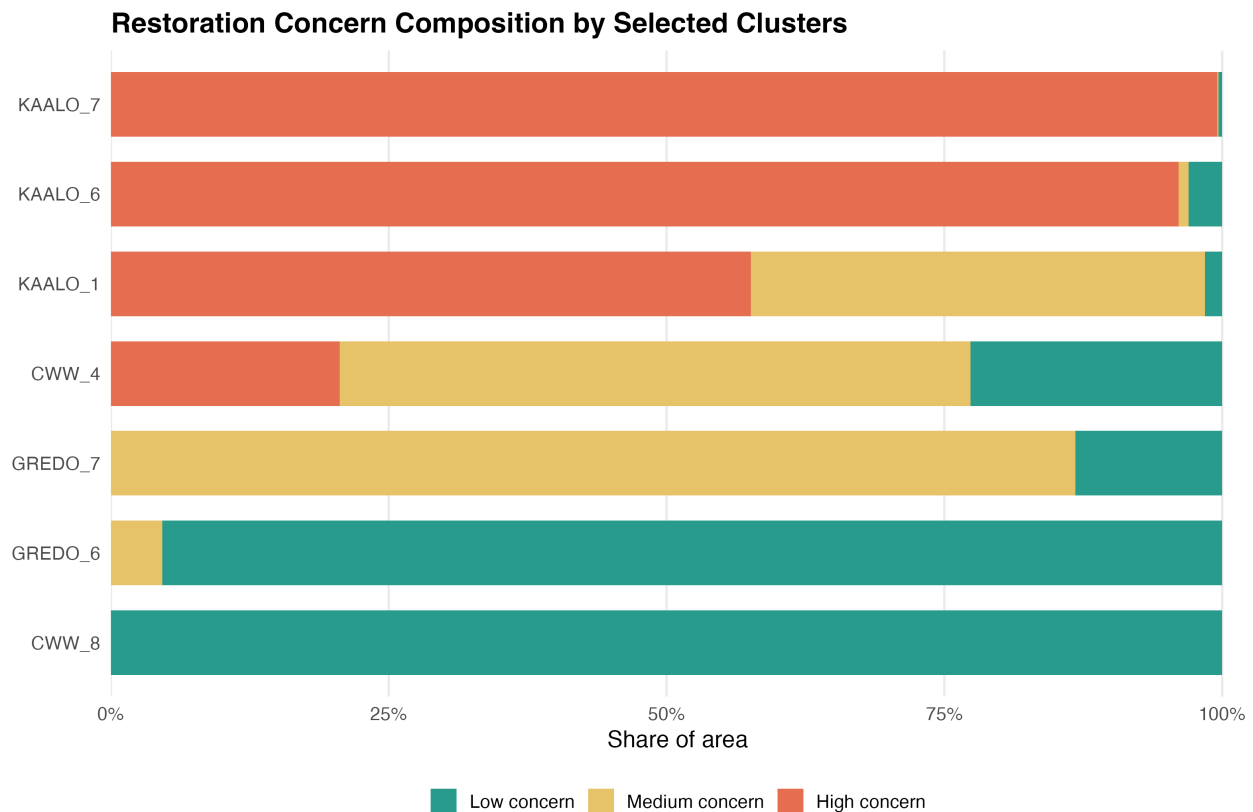


Figure 6: Share of restoration concern classes by selected project cluster.

The figure above shows the distribution of the three concern classes across the clusters. The results show that KAALO_7 in the Nugaal region has the highest percentage of area under high concern, while GREDO_7 and GREDO_6 in Bay region barely have any area under high concern. CWW_8 in Gedo region is completely under low concern.

The analysis can further be broken down into the watersheds and clusters contained within each watershed; this analysis is given in detail below.

6 Watershed and Cluster-Level Analysis

The graphs below compare indicator distributions at national, watershed and cluster scales. Under the high concern class, erosion and pH distributions are shifted to the right (higher values), indicating more severe erosion and less favorable pH conditions. In contrast, SOC, RDR50, and tree cover distributions are shifted left (lower values), reflecting reduced soil carbon, shallower effective rooting depth, and sparser tree cover in these areas.

Across the high concern class indicators, the watershed median is generally higher than the national median, except for SOC, where the watershed median is lower. This suggests that, relative to the country, the watershed experiences more intense erosion, more extreme pH, and greater root-depth restriction, lower soil organic carbon but has higher tree cover.

Medium concern classes show a roughly balanced (half-half) distribution, indicating mixed conditions with no strong skew toward extreme values. Low concern areas display a pattern nearly opposite to that of high concern areas; where high-concern distributions are skewed toward unfavourable values (e.g., high erosion/pH, low SOC/RDR50/tree cover), low-concern distributions are skewed toward more favourable values, and any overlap in direction covers a smaller area under the curve.

Across subsequent watersheds and clusters in the following sections, the distribution shapes under the three concern classes remain generally similar; only the area under the curve varies. This indicates that the same general patterns of indicator behavior (e.g., shifts toward higher erosion/pH and lower SOC/RDR50/tree cover in high-concern areas) hold across subregions, with differences mainly in the extent of each pattern rather than in its direction.

6.1 Watershed 1 and KAALO_6

The map below shows the restoration concern classes within Watershed 1 and highlights the KAALO_6 cluster contained within the watershed.

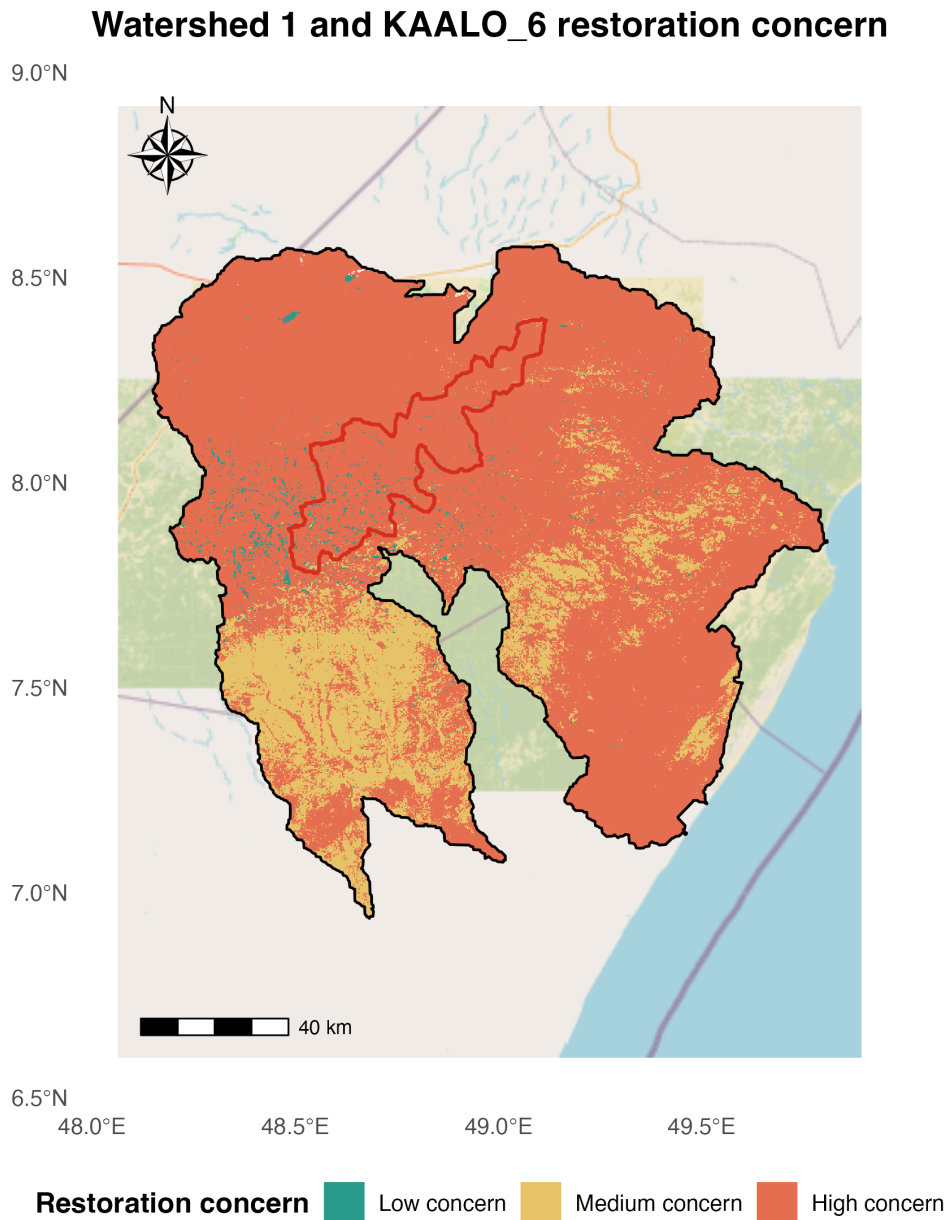


Figure 7: Restoration concern map for Watershed 1 and the KAALO_6 cluster.

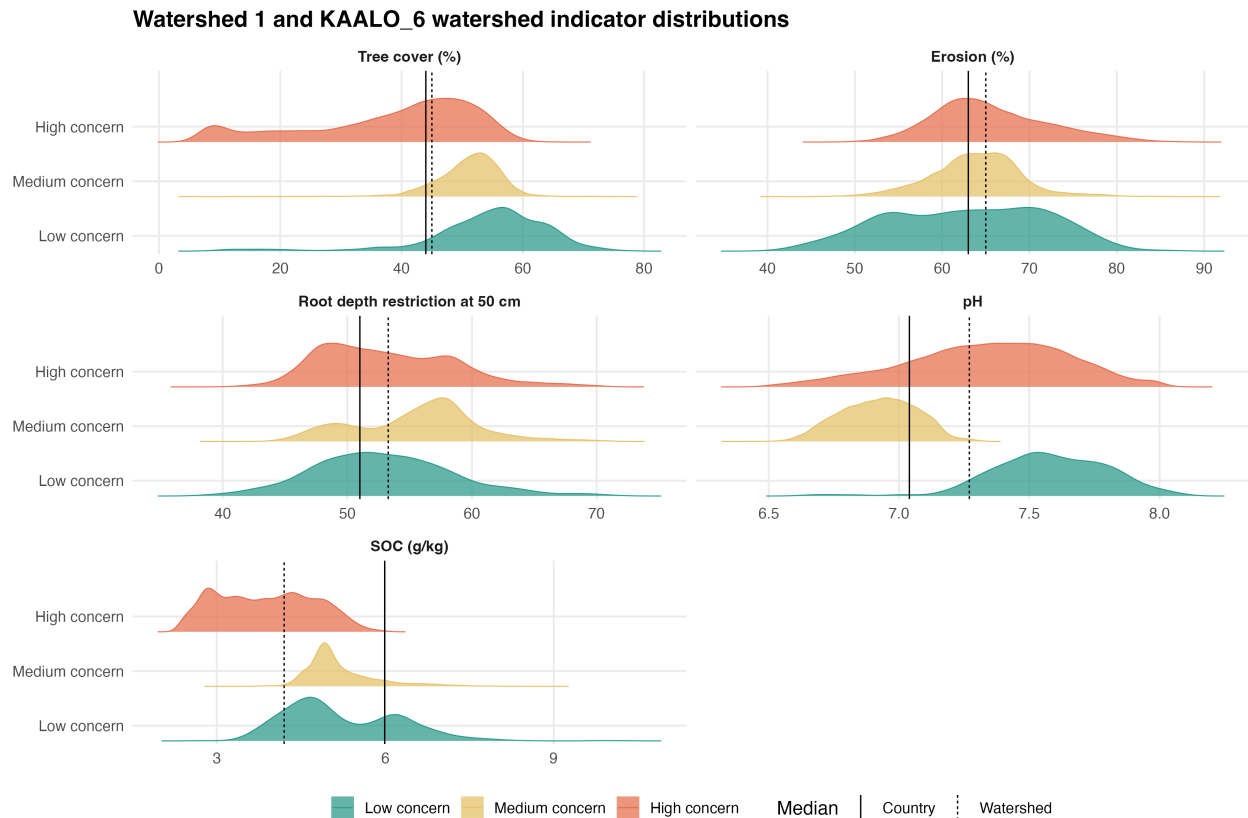


Figure 8: Watershed 1 indicator distributions by restoration concern class.

The watershed ridgeline plot shows the distribution of each indicator across the three concern classes, with median lines showing how Watershed 1 compares to the country median. Across the indicators, the watershed median is generally higher than the national median, except for SOC, where the watershed median is lower. This suggests that, relative to the country, Watershed 1 experiences more intense erosion, more extreme pH and greater root-depth restriction, with lower soil organic carbon but higher tree cover.

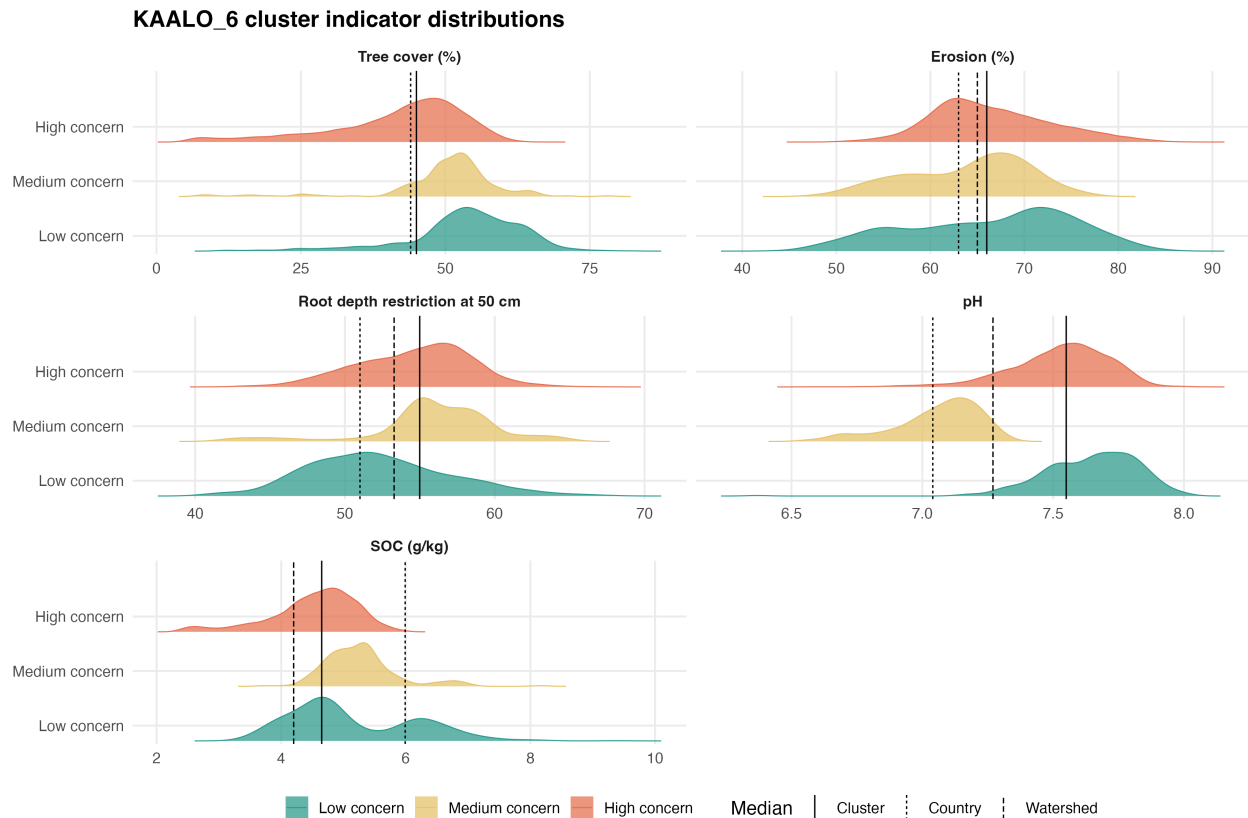


Figure 9: KAALO_6 indicator distributions by restoration concern class.

The KAALO_6 ridgeline plot compares the cluster median with the watershed and country medians. Across the indicators, the KAALO_6 median is generally higher than the national and watershed medians, except for SOC, where it is lower than the country median but higher than the watershed median.

6.2 Watershed 2 and KAALO_1

Watershed 2 contains the KAALO_1 cluster. The map below shows the restoration concern classes within Watershed 2 and highlights the KAALO_1 cluster contained within the watershed.

Watershed 2 and KAALO_1 restoration concern

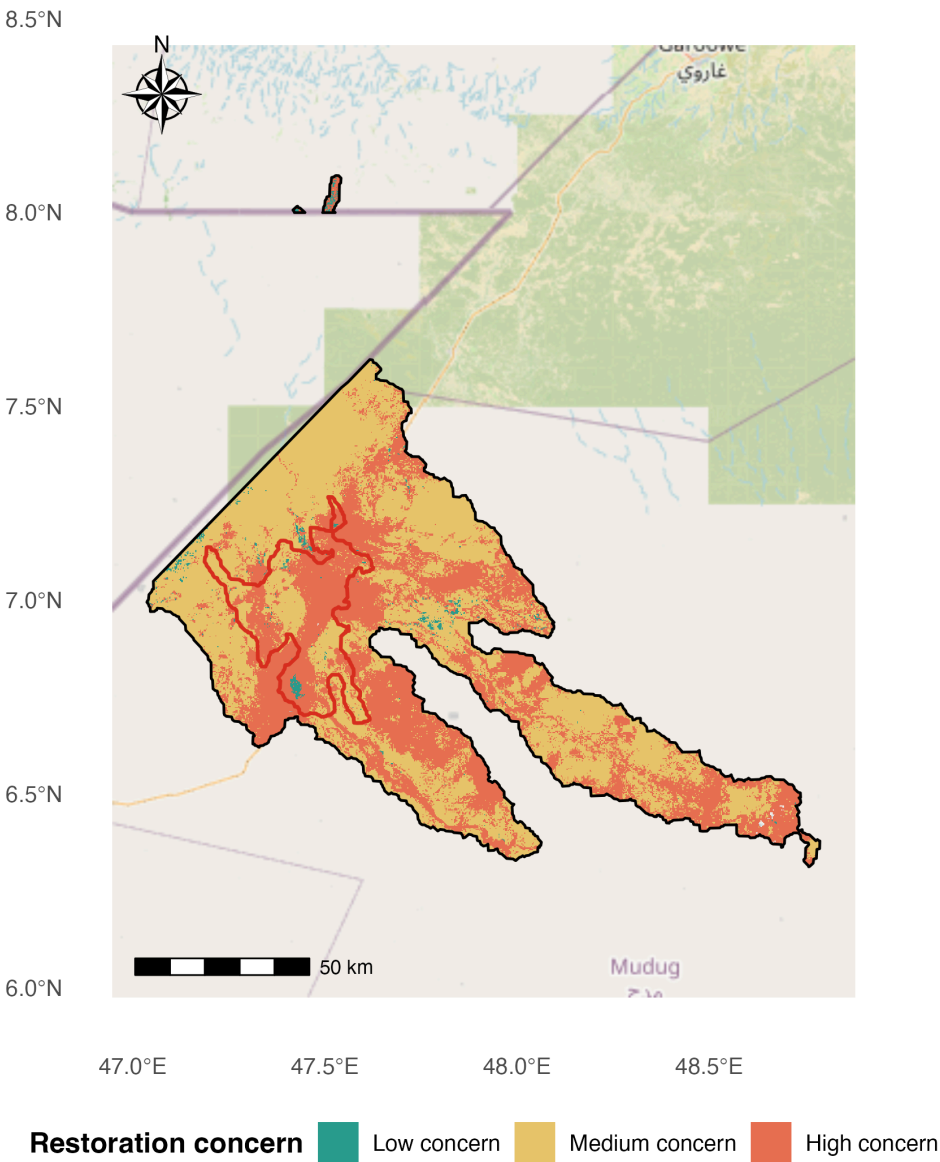


Figure 10: Restoration concern map for Watershed 2 and the KAALO_1 cluster.

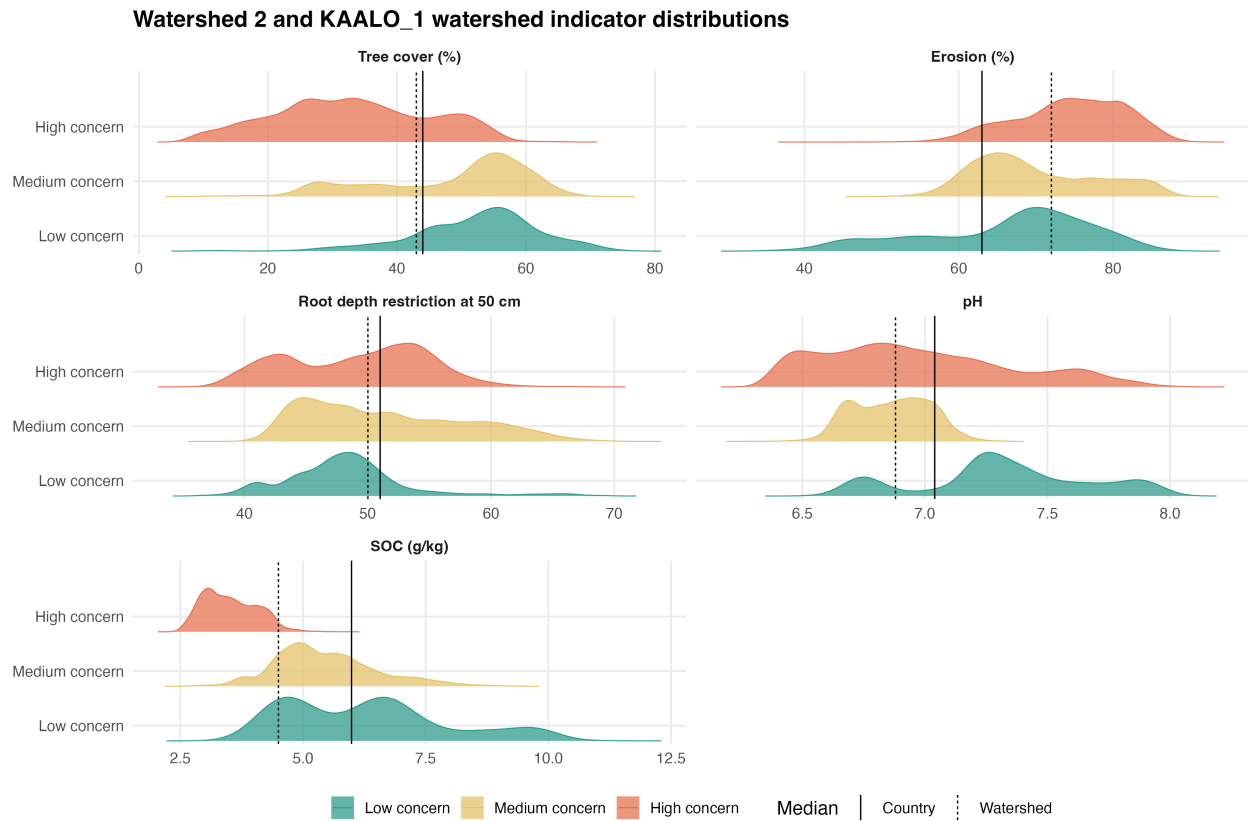


Figure 11: Watershed 2 indicator distributions by restoration concern class.

The watershed ridgeline plot shows how Watershed 2 compares to the national median across indicators and concern classes. Across the indicators, the watershed median is generally lower than the national median except for erosion. This suggests that, relative to the country, the watershed experiences higher degradation pressure due to higher erosion, lower SOC, lower tree cover and limited rooting depth.

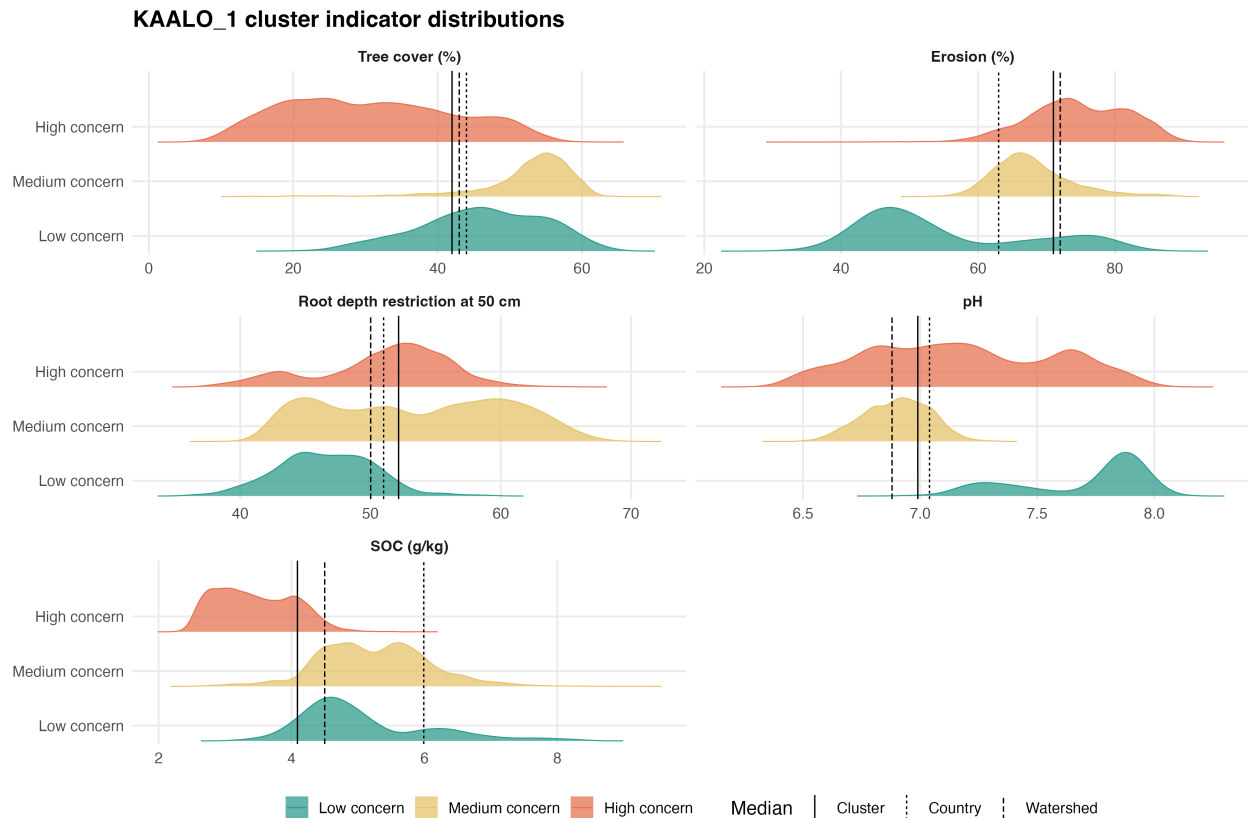


Figure 12: KAALO_1 indicator distributions by restoration concern class.

Across the indicators, the KAALO_1 median shows a mixed pattern where it is higher than the watershed and national medians for root depth restriction, lower than both for tree cover and SOC while lying in between the two for pH.

This pattern suggests that KAALO_1 experiences intermediate erosion pressure and pH conditions than either the broader watershed or the country median. At the same time, lower SOC and tree cover imply weaker vegetation structure and reduced SOC, pointing to conditions that may be more degraded or more vulnerable to further decline unless restoration specifically targets vegetation restoration and SOC accumulation.

6.3 Watershed 3 and CWW_4

Watershed 3 contains the CWW_4 cluster. The map below shows the restoration concern classes within Watershed 3 and highlights the CWW_4 cluster contained within the watershed.

Watershed 3 and CWW_4 restoration concern

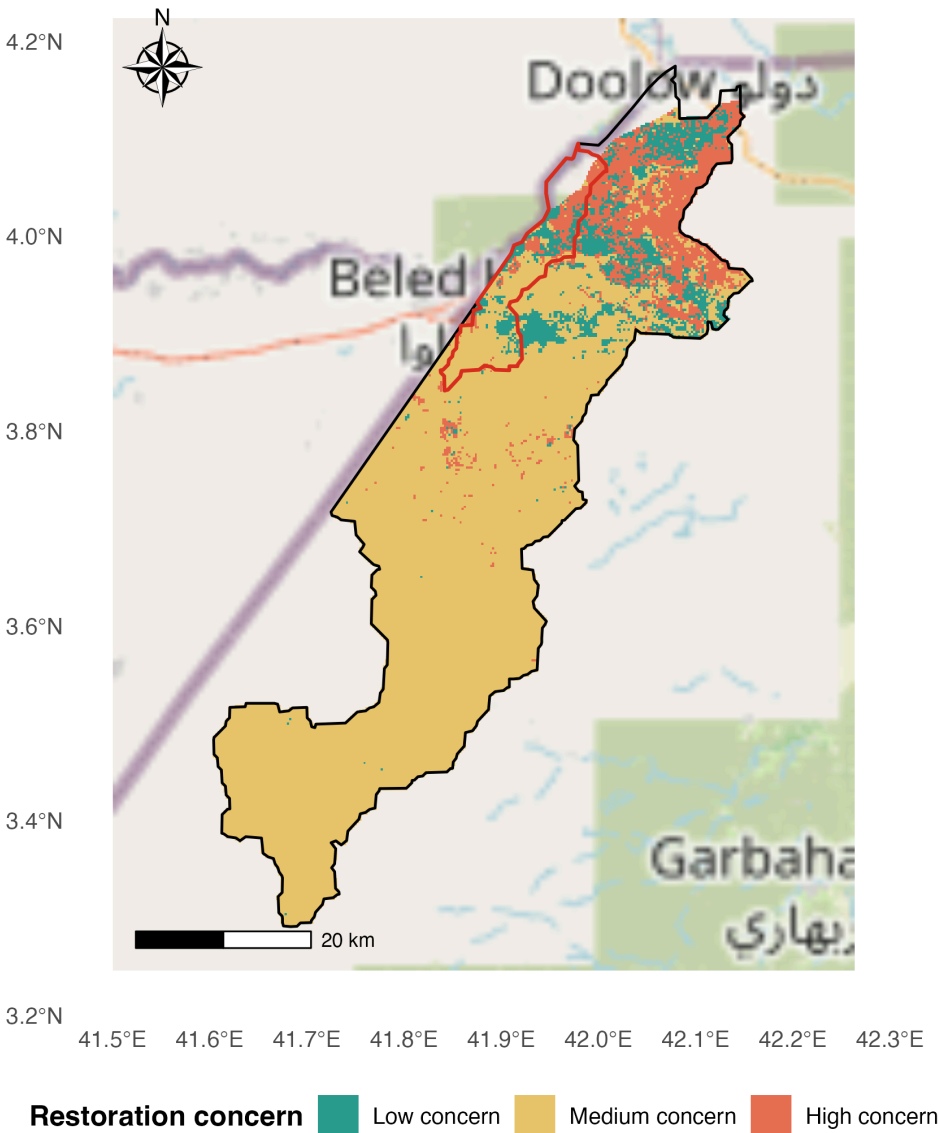


Figure 13: Restoration concern map for Watershed 3 and the CWW_4 cluster.

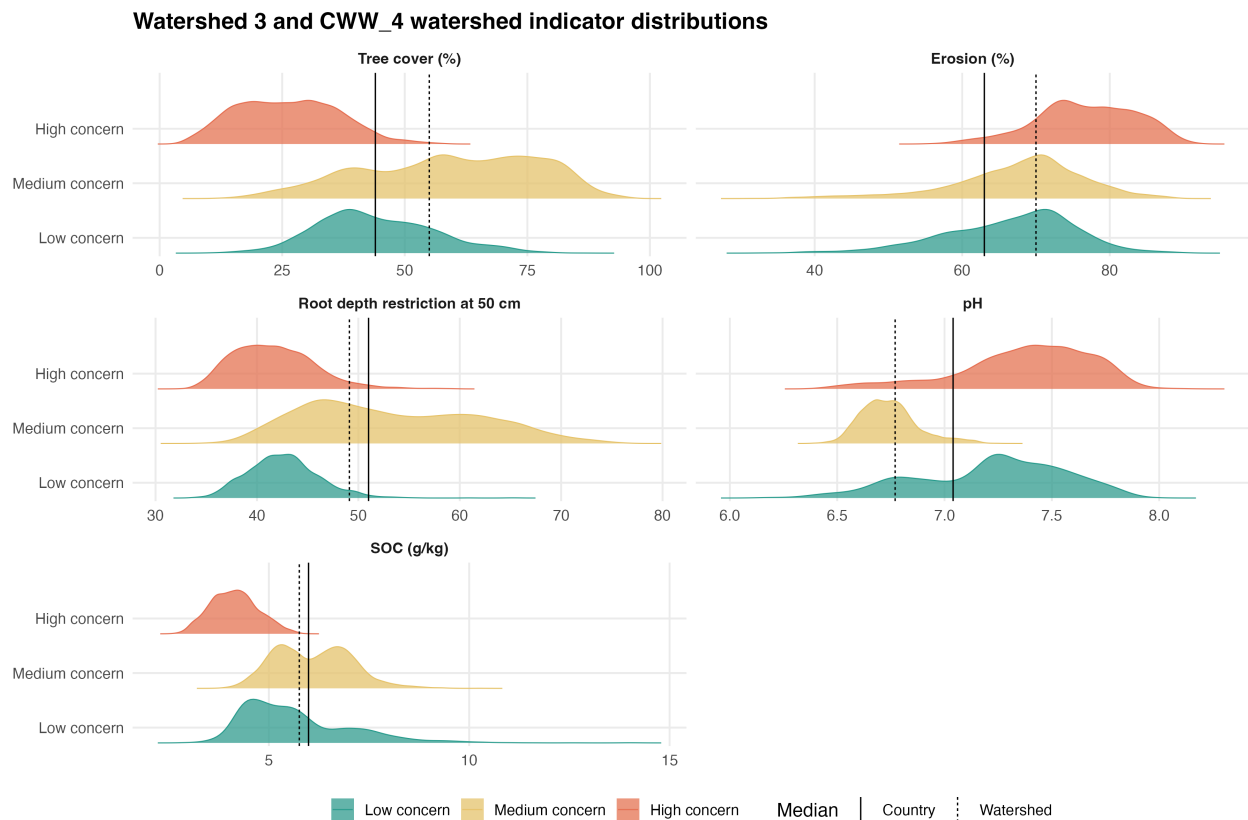


Figure 14: Watershed 3 indicator distributions by restoration concern class.

The watershed ridgeline plot shows how Watershed 3 compares to the national median across indicators and concern classes. Across the indicators, the watershed median is generally lower than the national median, except for erosion and tree cover, where the watershed median is higher. This indicates that the watershed experiences stronger erosion pressure and higher tree cover, but relatively lower pH, lower root-depth restriction and lower soil organic carbon compared with the national scale. This may signal that the restoration should focus on ground cover restoration, runoff control and gully stabilization since the tree cover is relatively high but erosion is as well.

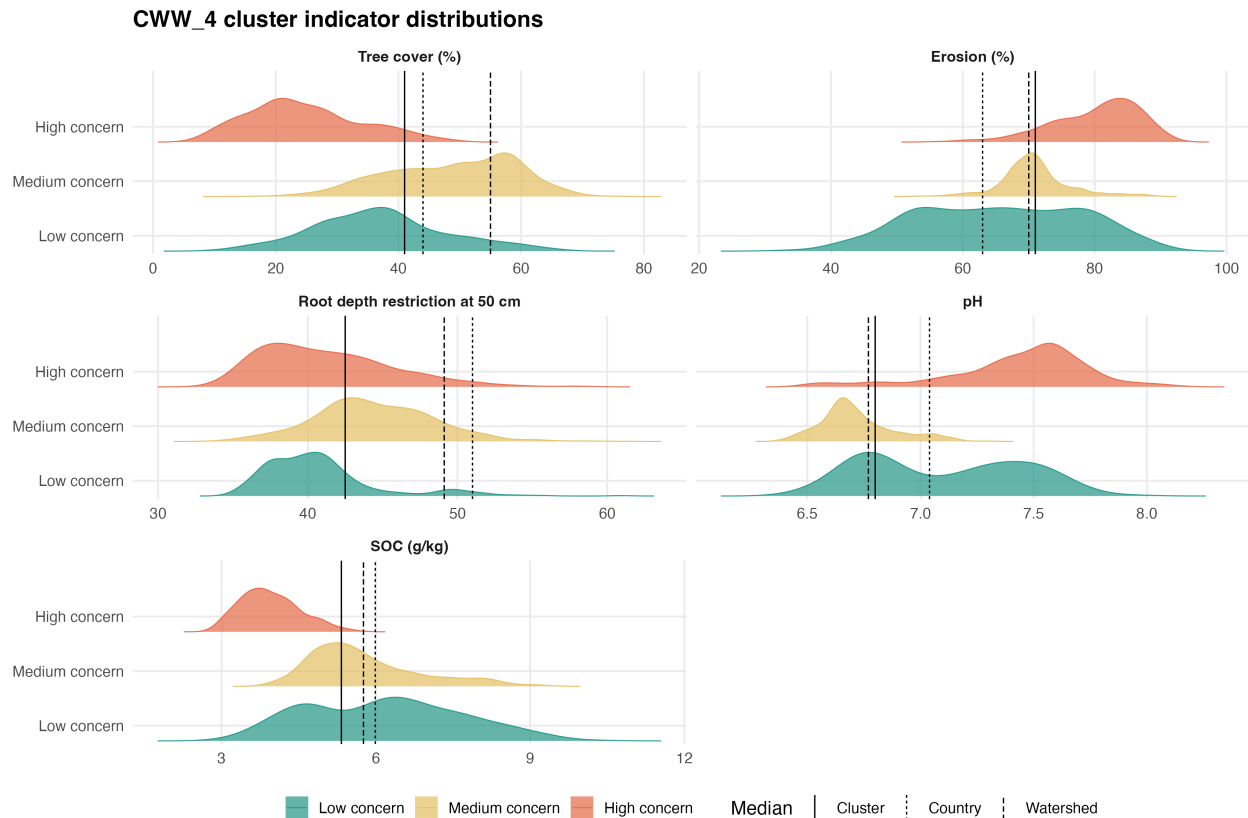


Figure 15: CWW_4 indicator distributions by restoration concern class.

Across the indicators, the CWW_4 median is lower than the country and watershed medians for root depth restriction, SOC and tree cover, but higher than both for erosion and intermediate for pH. These results suggest that the cluster faces stronger erosion pressure, more severe root depth restriction, and reduced vegetation and SOC relative to the country and watershed scales, while experiencing intermediate pH conditions.

6.4 Watersheds 5 and 6 with GREDO_6 and GREDO_7

Watersheds 5 and 6 contain the GREDO_6 and GREDO_7 clusters. The map below shows the restoration concern classes within Watersheds 5 and 6 and highlights the GREDO_6 and GREDO_7 clusters contained within these watersheds.

Watersheds 5 and 6 with GREDO_6 and GREDO_7 restoration concern

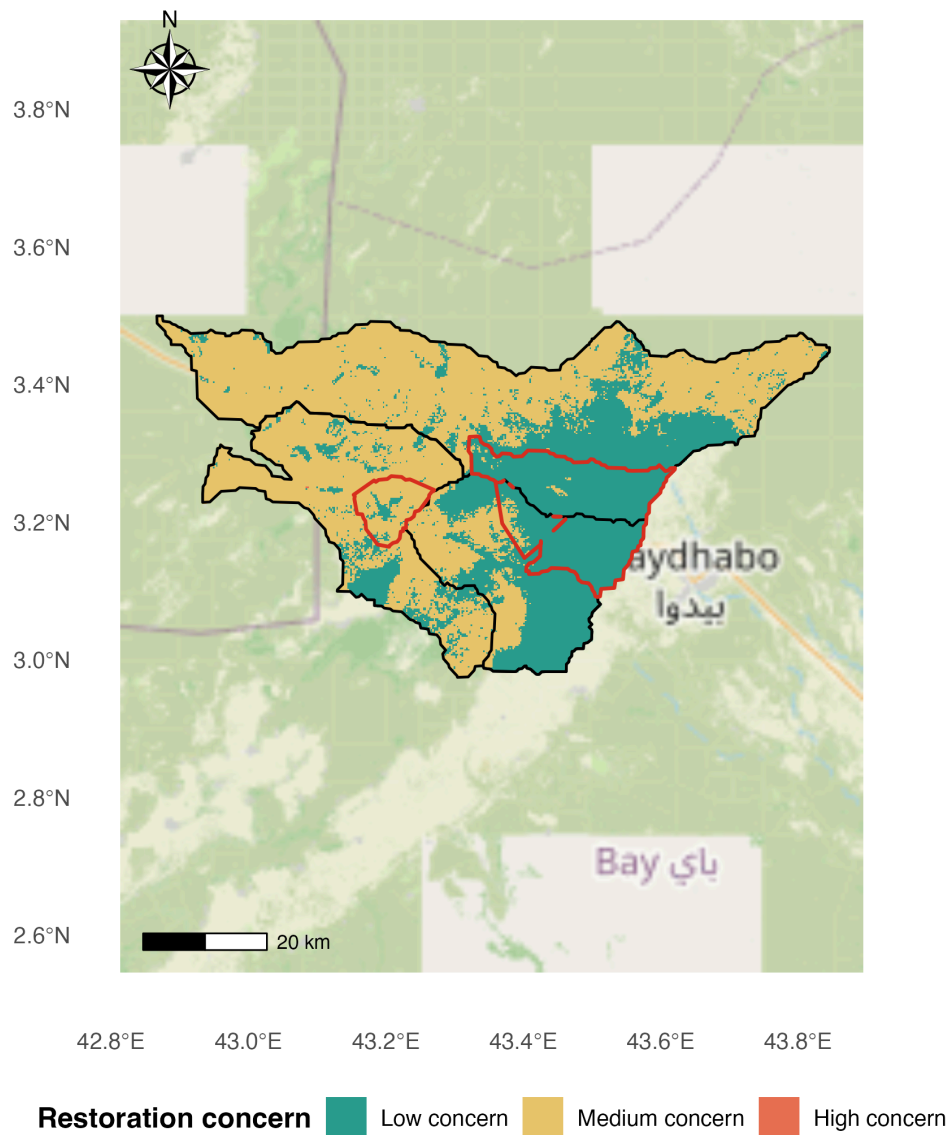


Figure 16: Restoration concern map for Watersheds 5 and 6 and the GREDO_6 and GREDO_7 clusters.



Figure 17: Watersheds 5 and 6 indicator distributions by restoration concern class.

The watershed ridgeline plot shows how Watersheds 5 and 6 compare to the national median across indicators and concern classes. Conditions are dominated by medium concern followed by low concern, while high concern is present only as a very small trace area. Based on the raster area summary, the combined high concern area across Watersheds 5 and 6 is about 0.31 km², or roughly 0.009% of the mapped area in these two watersheds. This is why the high concern class appears in the ridgeline plot, but is barely visible in the area-based composition graph and map.

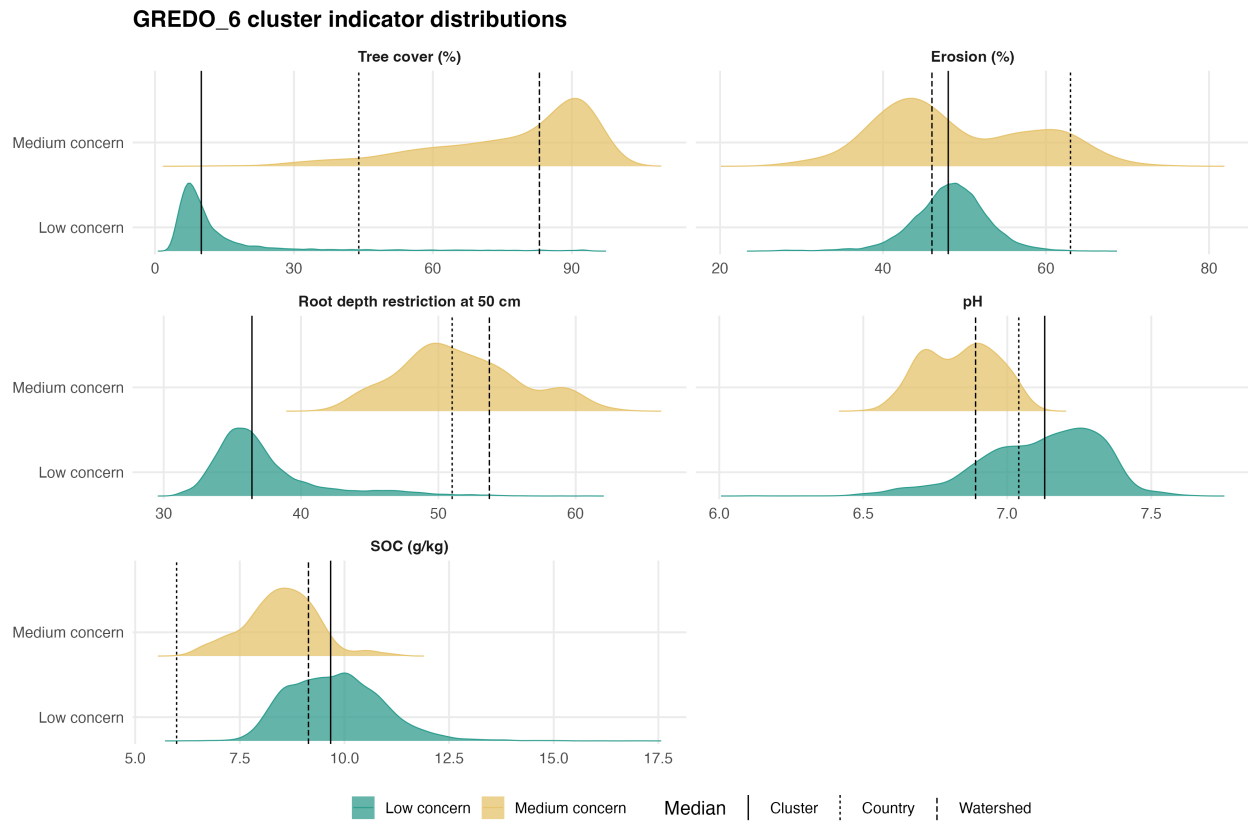


Figure 18: GREDO_6 indicator distributions by restoration concern class.

For GREDO_6, the median is lower than the country and watershed medians for root depth restriction and tree cover, but higher than both for pH and SOC. Erosion is intermediate, with the country erosion median higher than the cluster median. This suggests shallower effective rooting depth and more sparse tree cover than the broader regions, but more favourable pH, higher SOC and lower erosion compared with the national scale.

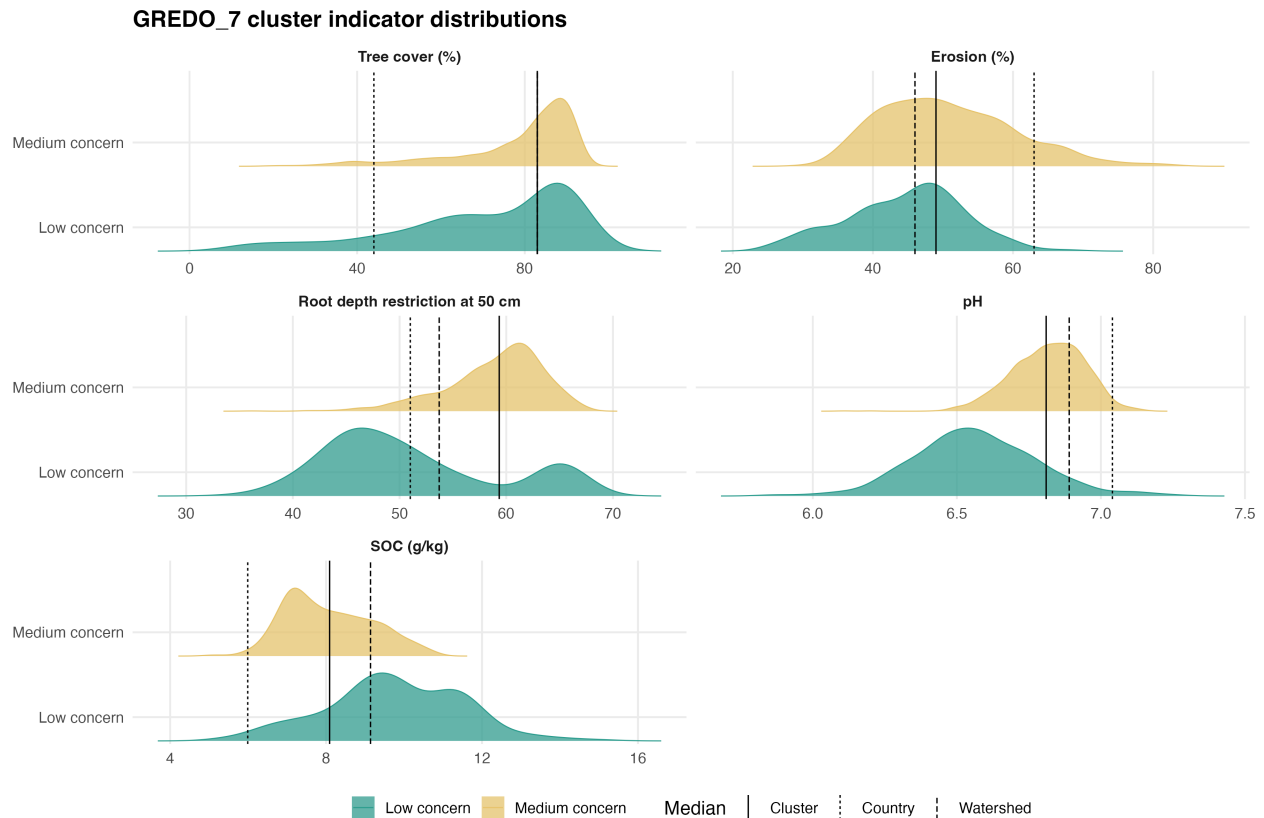


Figure 19: GREDO_7 indicator distributions by restoration concern class.

For GREDO_7, the median is lower than the country and watershed medians for pH, higher than both for root depth restriction and tree cover, and intermediate for erosion and SOC. This suggests less favourable pH conditions but deeper effective rooting depth and denser tree cover than both broader scales. Erosion pressure is moderately lower than the country median but slightly above the watershed median, while SOC is better than the country overall but not as high as the watershed median.

6.5 Watershed 7 and CWW_8

Watershed 7 contains the CWW_8 cluster. The map below shows the restoration concern classes within Watershed 7 and highlights the CWW_8 cluster contained within the watershed.

Watershed 7 and CWW_8 restoration concern

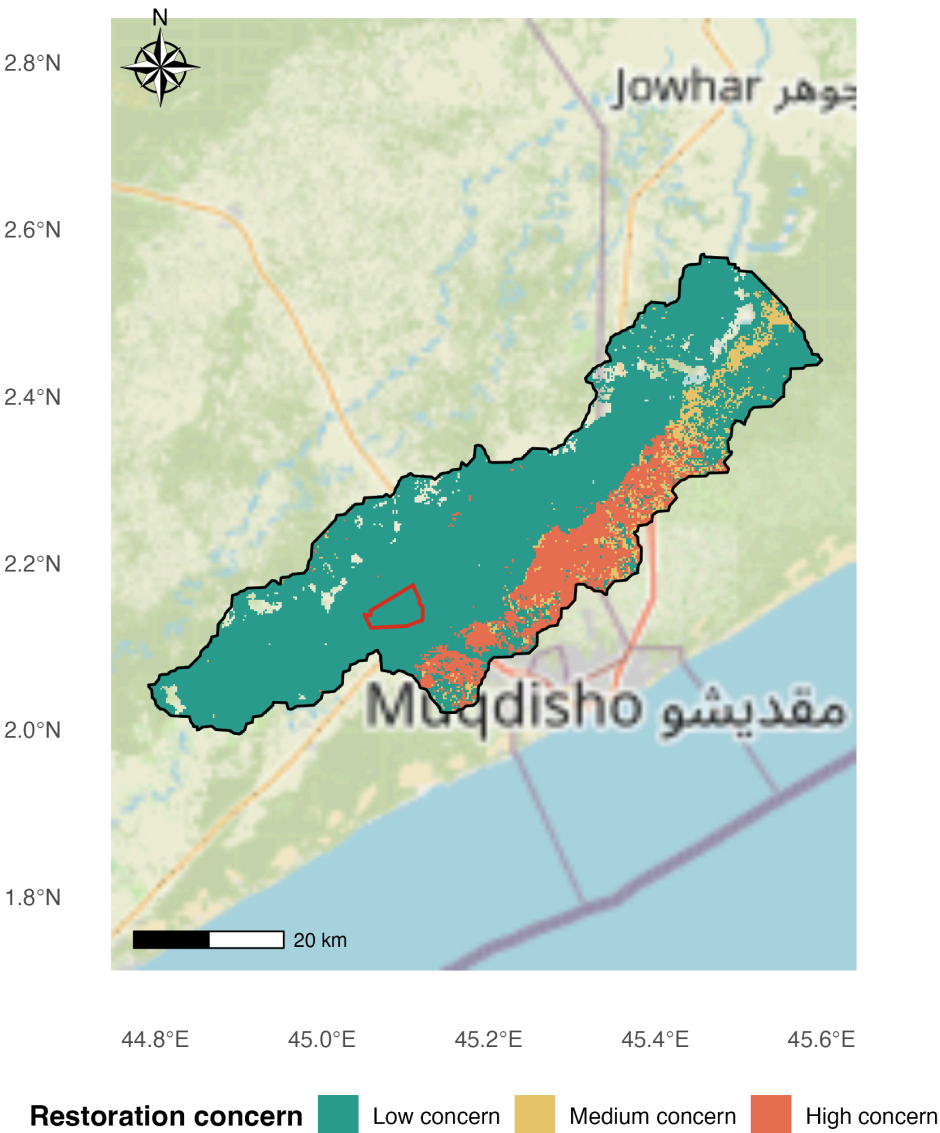


Figure 20: Restoration concern map for Watershed 7 and the CWW_8 cluster.

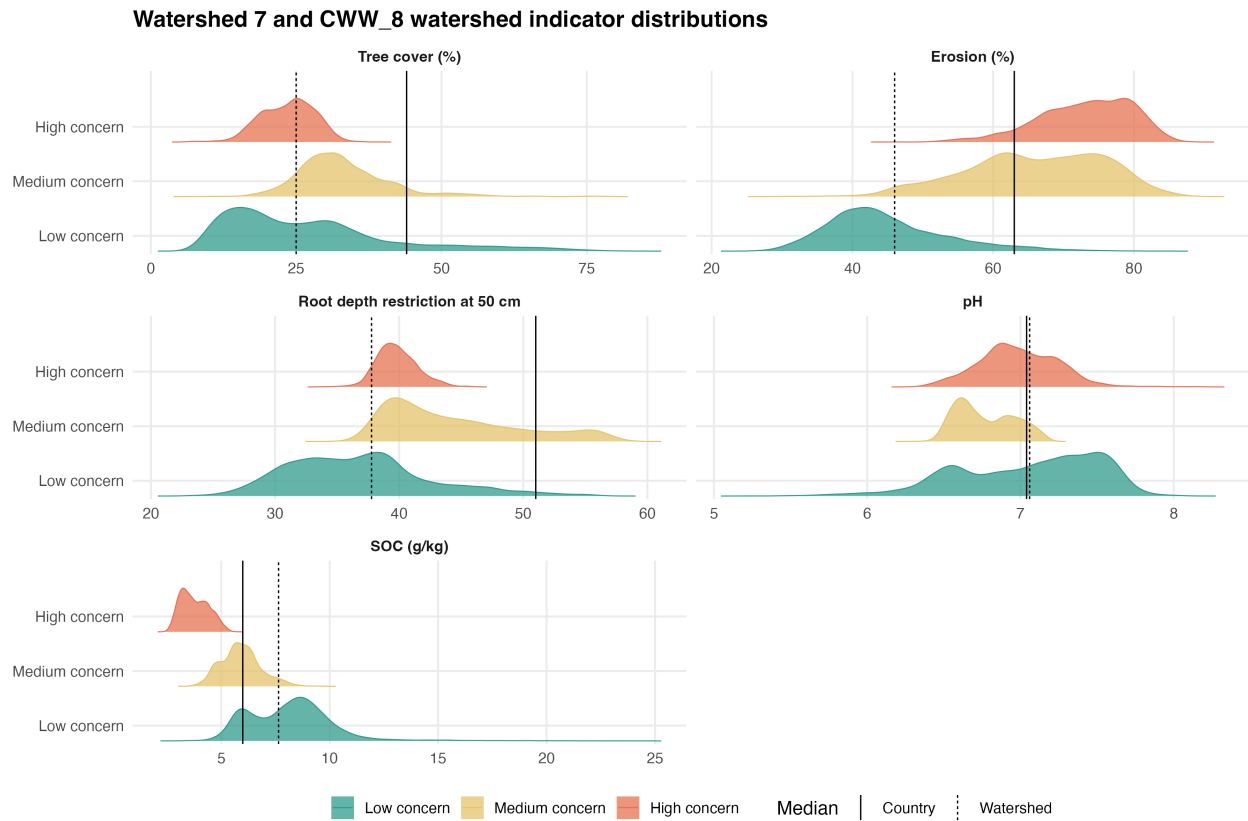


Figure 21: Watershed 7 indicator distributions by restoration concern class.

The watershed ridgeline plot shows how Watershed 7 compares to the national median across indicators and concern classes. Across the indicators, the watershed median is generally lower than the country median, except for SOC and pH, where the watershed median is higher. This suggests that the watershed has higher soil organic carbon and slightly higher pH compared to the national scale, while erosion, root depth restriction and tree cover tend to be lower.

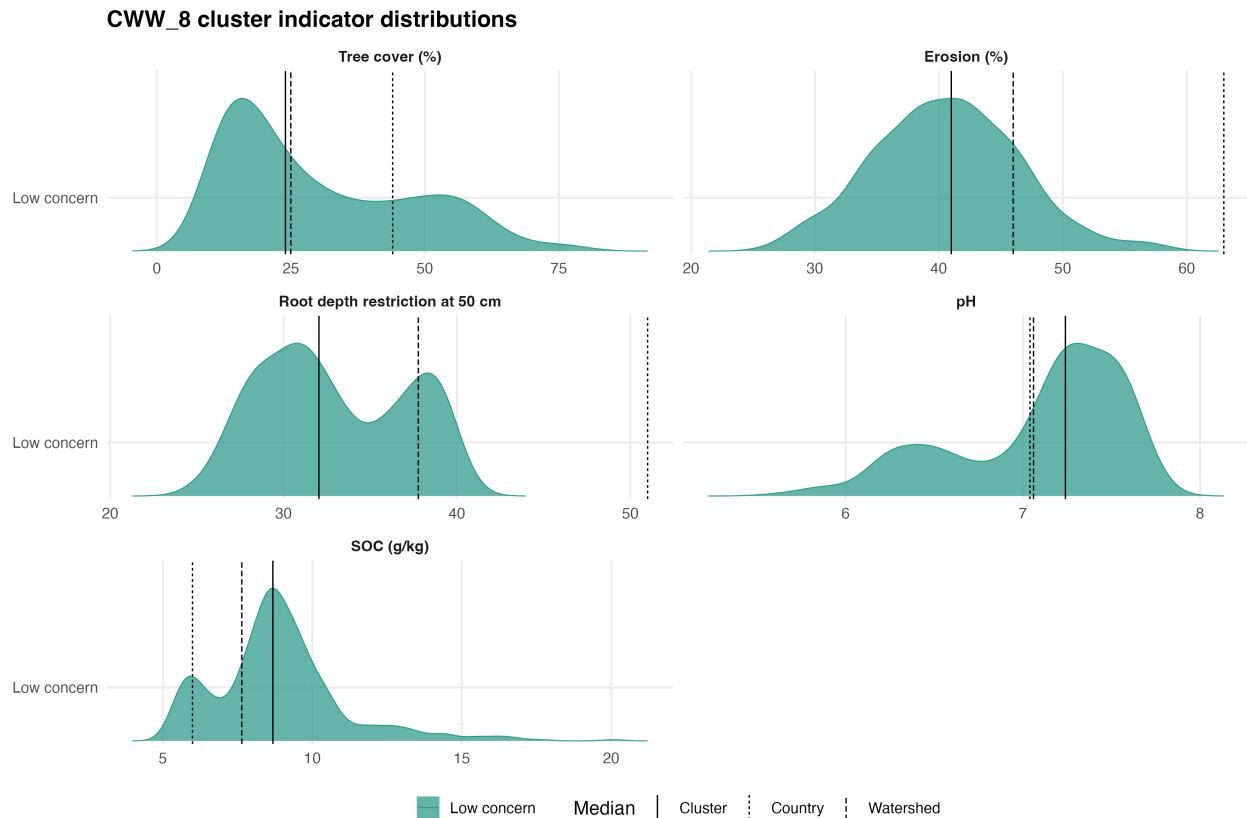


Figure 22: CWW_8 indicator distributions by restoration concern class.

Across indicators, the CWW_8 median is lower than the country and watershed medians for erosion, tree cover and root depth restriction, but higher for pH and SOC. The cluster falls entirely within the low-concern class, so restoration efforts should focus on conservation and enhancing vegetation cover to maintain and improve existing conditions.

6.6 Watershed 8 and KAALO_7

Watershed 8 contains the KAALO_7 cluster. The map below shows the restoration concern classes within Watershed 8 and highlights the KAALO_7 cluster contained within the watershed.

Watershed 8 and KAALO_7 restoration concern

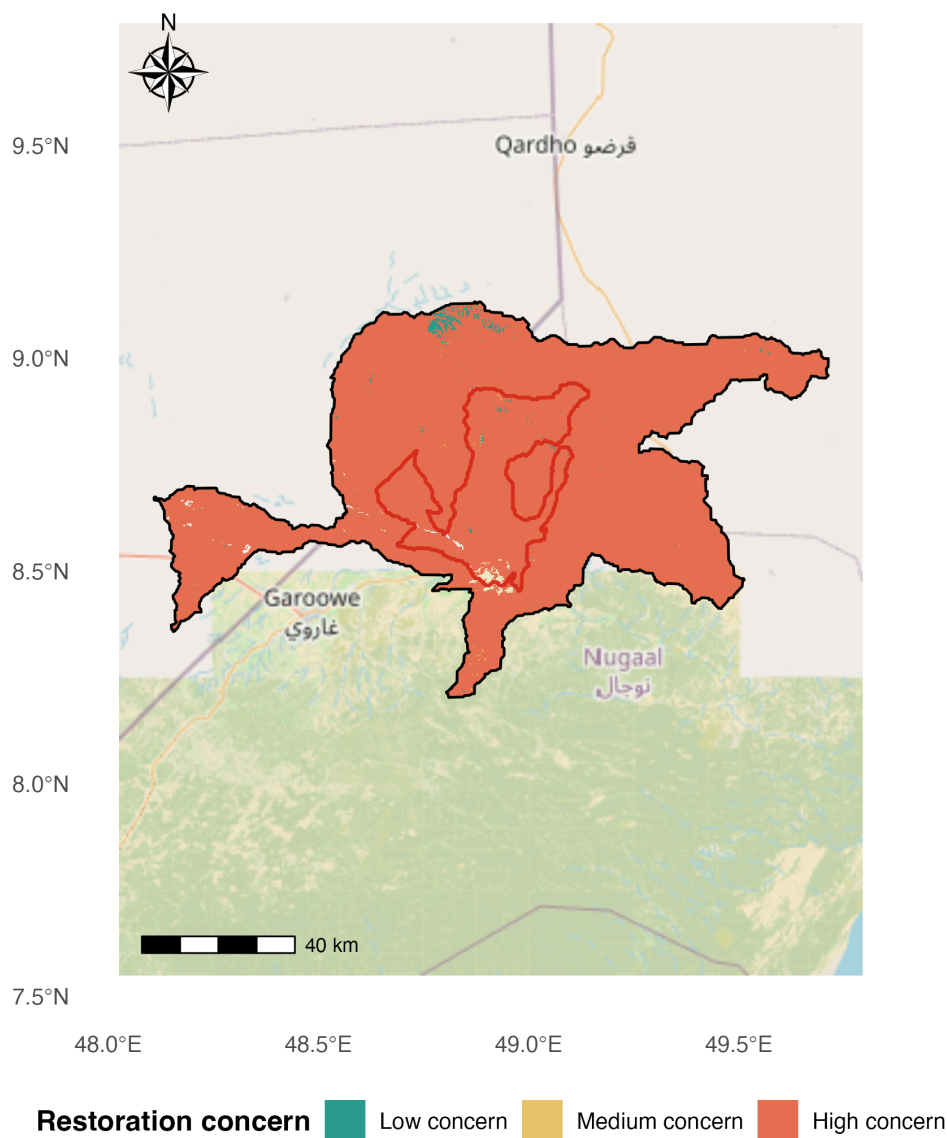


Figure 23: Restoration concern map for Watershed 8 and the KAALO_7 cluster.

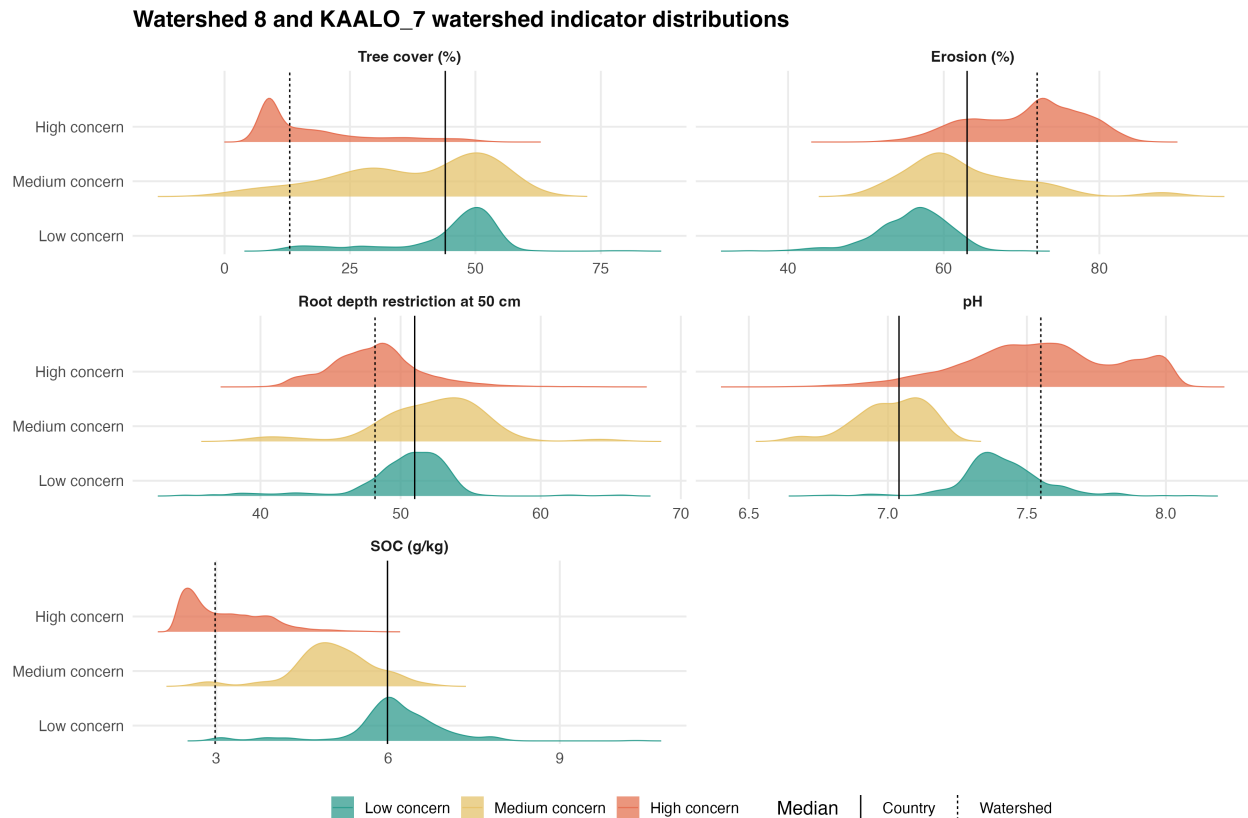


Figure 24: Watershed 8 indicator distributions by restoration concern class.

The watershed ridgeline plot shows how Watershed 8 compares to the national median across indicators and concern classes. The watershed and cluster are dominated by high concern as shown in the map as well as the distribution graphs where we see the high concern class in region being highly skewed in the negative directions for the indicators such as to the left for tree cover, root depth restriction and SOC; to the right for erosion and pH. This observation implies that, relative to the country, the watershed experiences more intense degradation across key indicators.

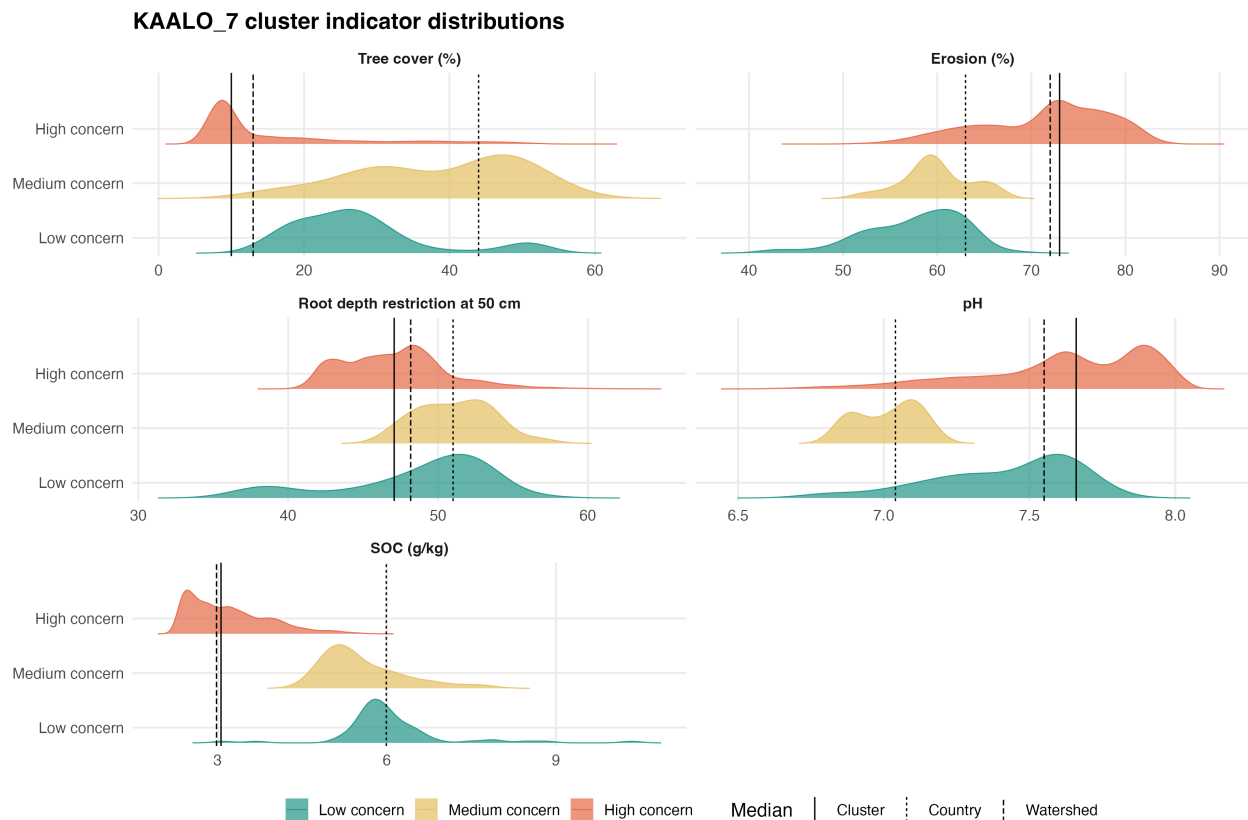


Figure 25: KAALO_7 indicator distributions by restoration concern class.

For KAALO_7, the pattern closely matches its parent watershed. The median is lower than the national median for root depth restriction and tree cover, close to the watershed median for SOC, and higher for pH and erosion. This indicates that the cluster is generally more degraded than the broader national scale. Restoration efforts in KAALO_7 should prioritize erosion control, soil conservation and increasing vegetation cover through interventions such as contour bunds, mulching and improved grazing management. Limited effective rooting depth may hinder the establishment of deep-rooted species; where feasible, soil decompaction could be considered before planting, or shallow-rooted, stress-tolerant species should be prioritized.

7 Restoration Recommendations

Low concern areas generally exhibit favourable ecosystem conditions, including high tree cover, moderate SOC, nearly neutral pH and relatively high RDR50 values. Intensive restoration is unlikely to be required, although moderate to high erosion levels indicate that soil and water conservation measures could help prevent further degradation. Restoration should focus on protection, maintenance and erosion control while preserving existing vegetation.

Medium concern areas exhibit mixed ecosystem conditions, characterized by moderate tree cover, relatively high SOC, slightly alkaline soils and intermediate root-depth restriction. The comparatively favourable soil conditions suggest that vegetation restoration remains feasible. Targeted tree planting, grass reseeding and improved grazing management may be appropriate where vegetation cover is declining or sparse. Localized soil and water conservation measures should also be considered, particularly in areas experiencing moderate or high erosion.

High concern areas show the strongest indicators of degradation, including very low tree cover, high erosion, very low SOC, alkaline soil conditions and relatively low RDR50 values. Restoration will likely require more active, sustained and resource-intensive intervention. Soil and water conservation measures such as halfmoons, contour bunds and terraces should be prioritized to reduce erosion and improve water retention before or along-

side vegetation restoration. In severely eroded locations, physical structures such as check dams or gabions may also be needed to stabilize channels and control sediment movement. Grass reseeding and targeted tree establishment can help increase vegetation cover and gradually rebuild SOC.

8 Conclusion

This report presents land health assessments for Somalia based on five biophysical indicators derived from the Land Degradation Surveillance Framework (LDSF). Model based clustering grouped the sampled indicator values into three restoration effort classes, which are also interpreted as levels of concern for land degradation. Of the mapped area, about 37% falls within medium and 34% within high restoration effort, together corresponding to roughly 408,155 km², or 71% of the country classified here as medium to high concern. Low effort (least concern) areas account for the remaining 29%.

Variable importance from the class specific random forest models shows that soil organic carbon is the strongest predictor of high restoration effort, reflecting the clear separation of SOC values between high and medium or low effort areas. Soil pH is particularly important for distinguishing the medium effort class, while erosion, although high overall, contributes relatively little to separating the classes because its distributions overlap strongly across them. The classification is therefore driven mainly by soil condition rather than by erosion alone, although the interactions between the two are important. The generally low SOC levels across much of the country underline the need for interventions that rebuild soil health.

The resulting maps and probability surfaces provide a spatially explicit basis for prioritizing restoration interventions in Somalia. High effort areas are where investments such as soil and water conservation combined with vegetation re-establishment are most urgently needed, while low effort areas can serve as reference landscapes for sustainable land management. The same workflow can be repeated at watershed and cluster level, and updated as new ground data become available.

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